



ELTON

Electrical system & wiring

Wiring diagrams — service manual

PDM25 V2 · VW LT31 1976 · JSN 398

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1 Introduction

This is the physical wiring and service manual for the conversion of a VW LT31 (1976, JSN 398) to a Hardwire PDM25 V2 + Raspberry Pi dashboard. Each diagram shows the PDM connector on the left, the cable in the middle (cross-section + actual color) and the consumer on the right.

Conventions: The connectors' title bar carries the housing's physical color (A grey, B black, C green, D brown). Cross-sections are given in mm². Ground is shown as a GND pin on the consumer rather than as a drawn cable. Wire colors reflect the wiring as physically inspected in the vehicle.

2 Connectors

2.1 Connector A — Inputs (GREY)

All input signals: levers, switches and senders. Switch-to-ground unless otherwise noted. Grey housing. Pin number on the left, cable in the middle (cross-section + actual color), consumer on the right.

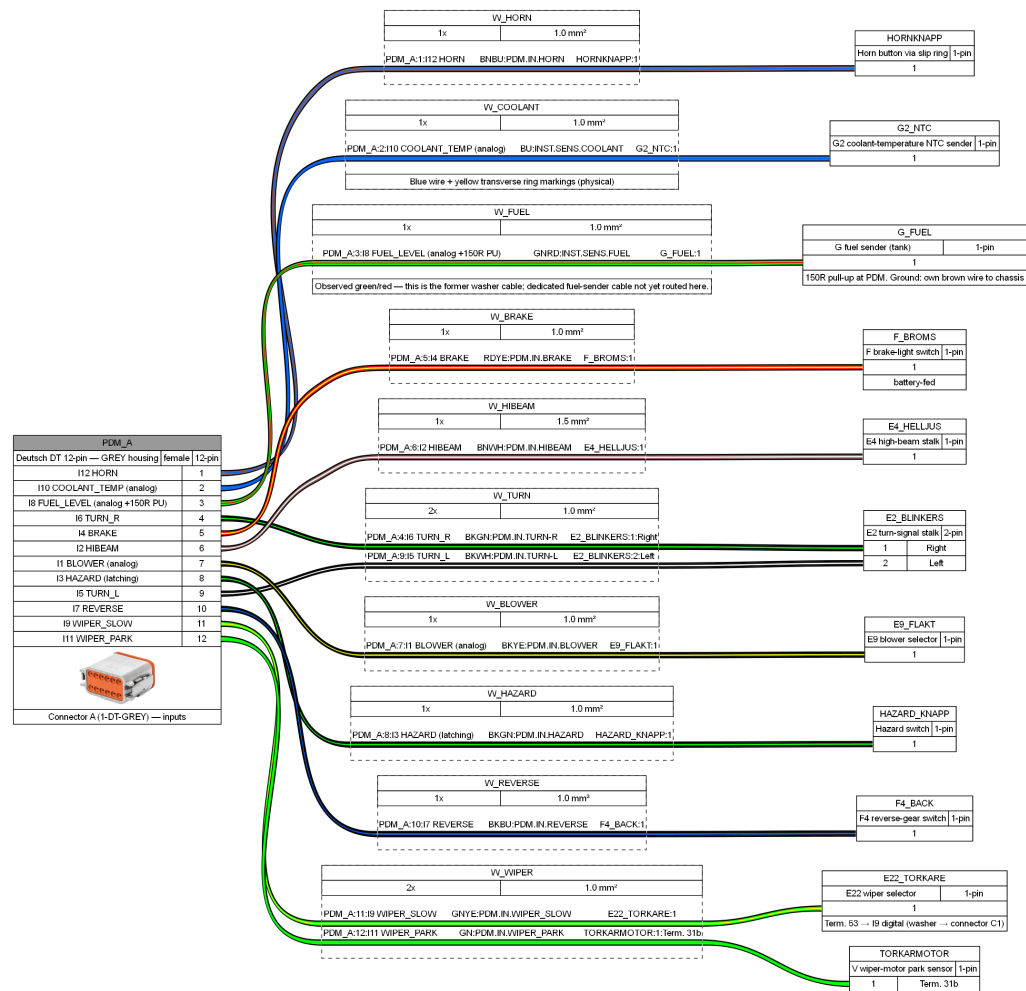


Diagram: wireviz_v8.0/1_kontakter/kontakt_A_inputs

2.2 Connector B — Outputs (BLACK)

Outputs: low beam, left turn signal, horn 1, wiper low, blower, washer pump, starter motor, reversing light, USB.

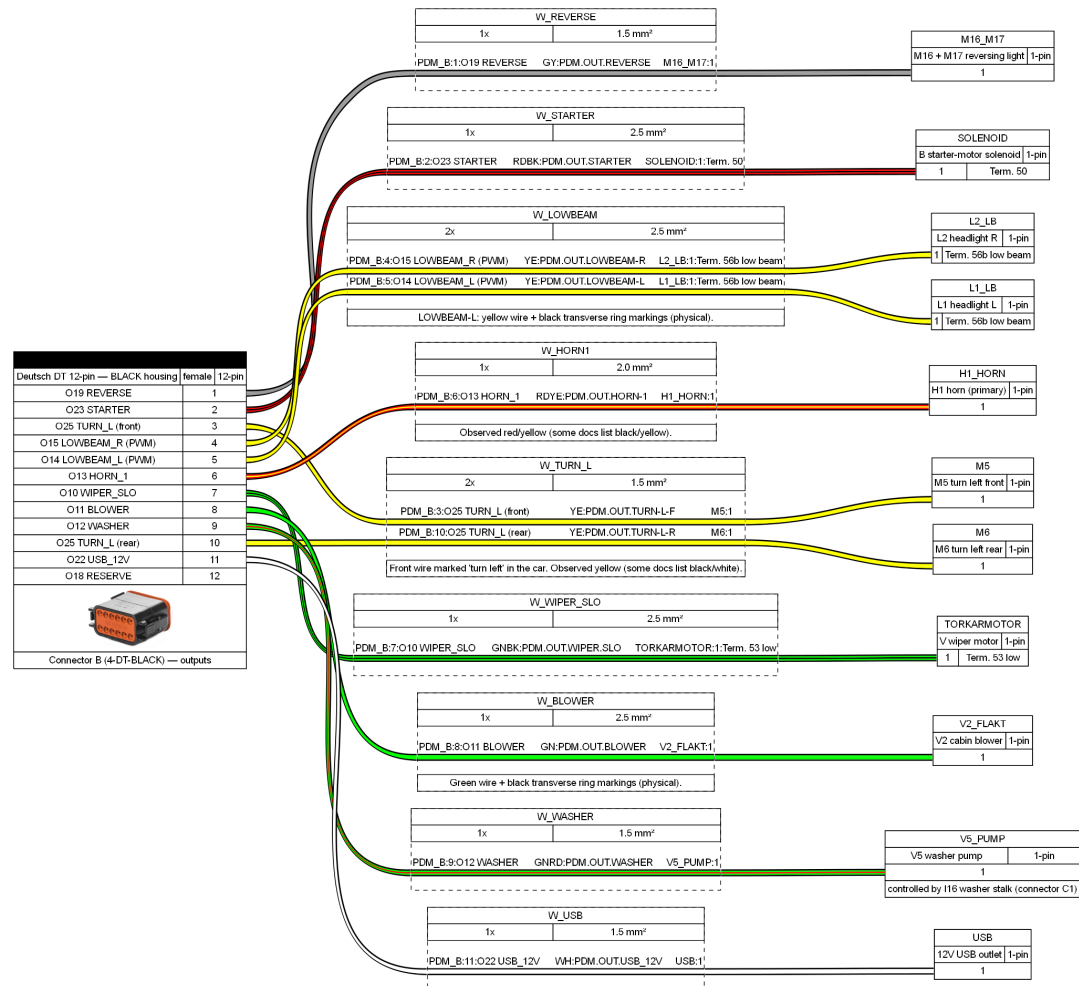


Diagram: wireviz_v8.0/1_kontakter/kontakt_B_outputs

2.3 Connector C — Power / CAN / sensors (GREEN)

Control feed (C3), chassis ground, CAN to the Pi, ignition/cranking, ignition coil, alternator field excitation, signal busbar and the 5V reference distribution point.

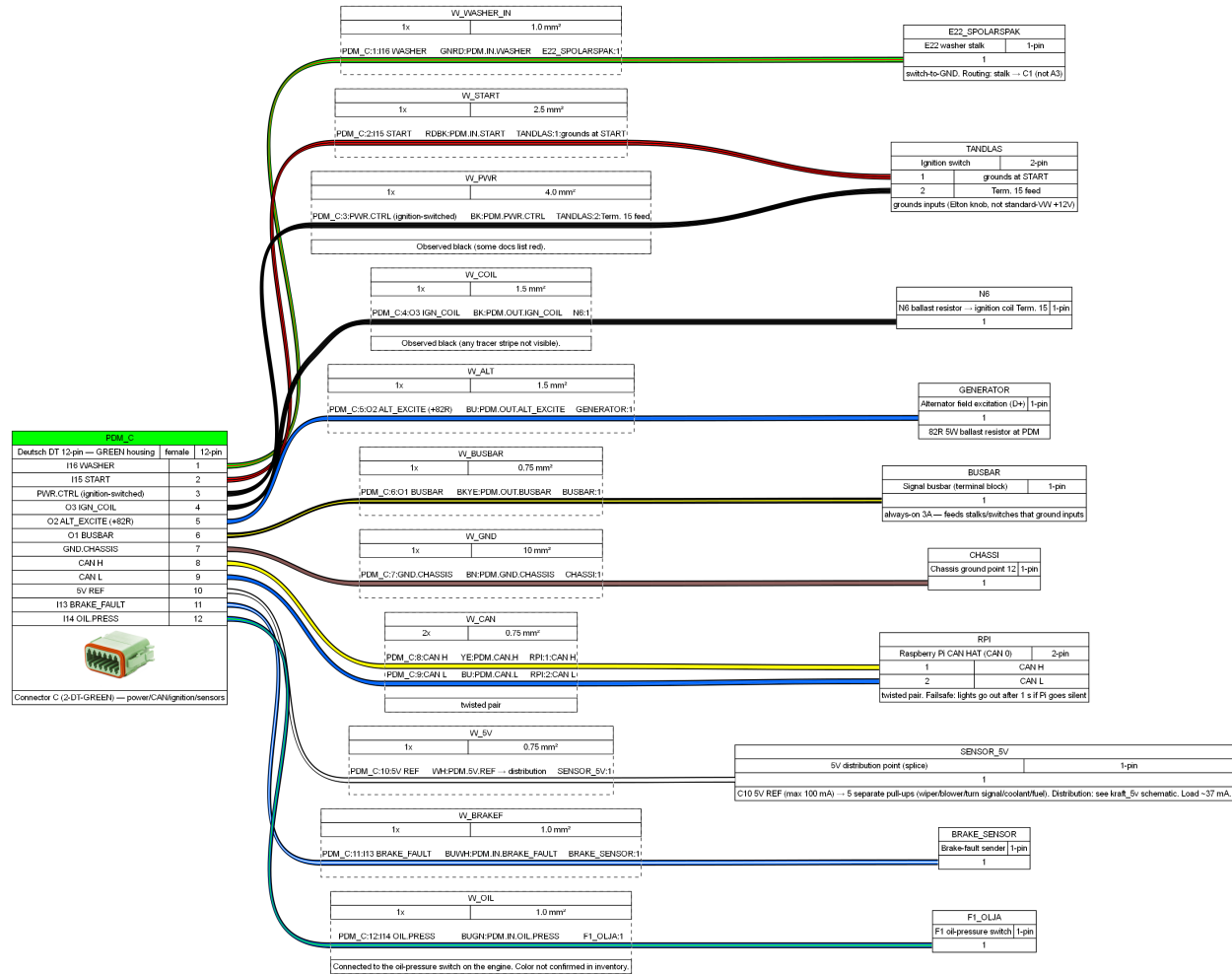


Diagram: wireviz_v8.0/1_kontakter/kontakt_C_kraft_can

2.4 Connector D — Outputs (BROWN)

Outputs: high beam, parking light, brake light, right turn signal, fuel pump (2 parallel to the same pin), wiper high, horn 2.

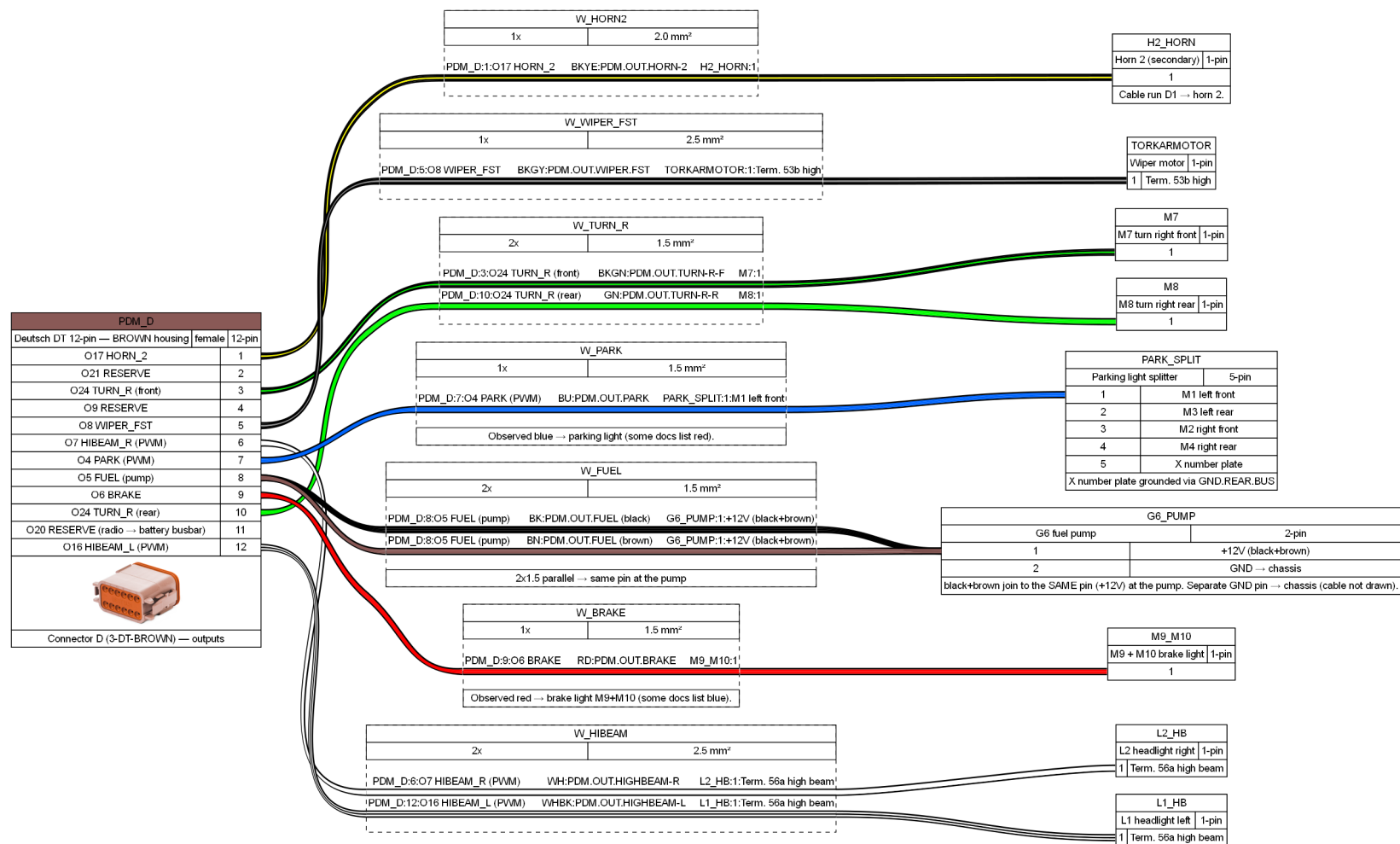


Diagram: wireviz_v8.0/1_kontakter/kontakt_D_outputs

3 Power supply

3.1 +12V power busbar (always-on loads)

The busbar (battery-fed via a 10A ceramic fuse) supplies the always-on loads: radio and Raspberry Pi. The PDM control feed C3 has its own graph (next page).

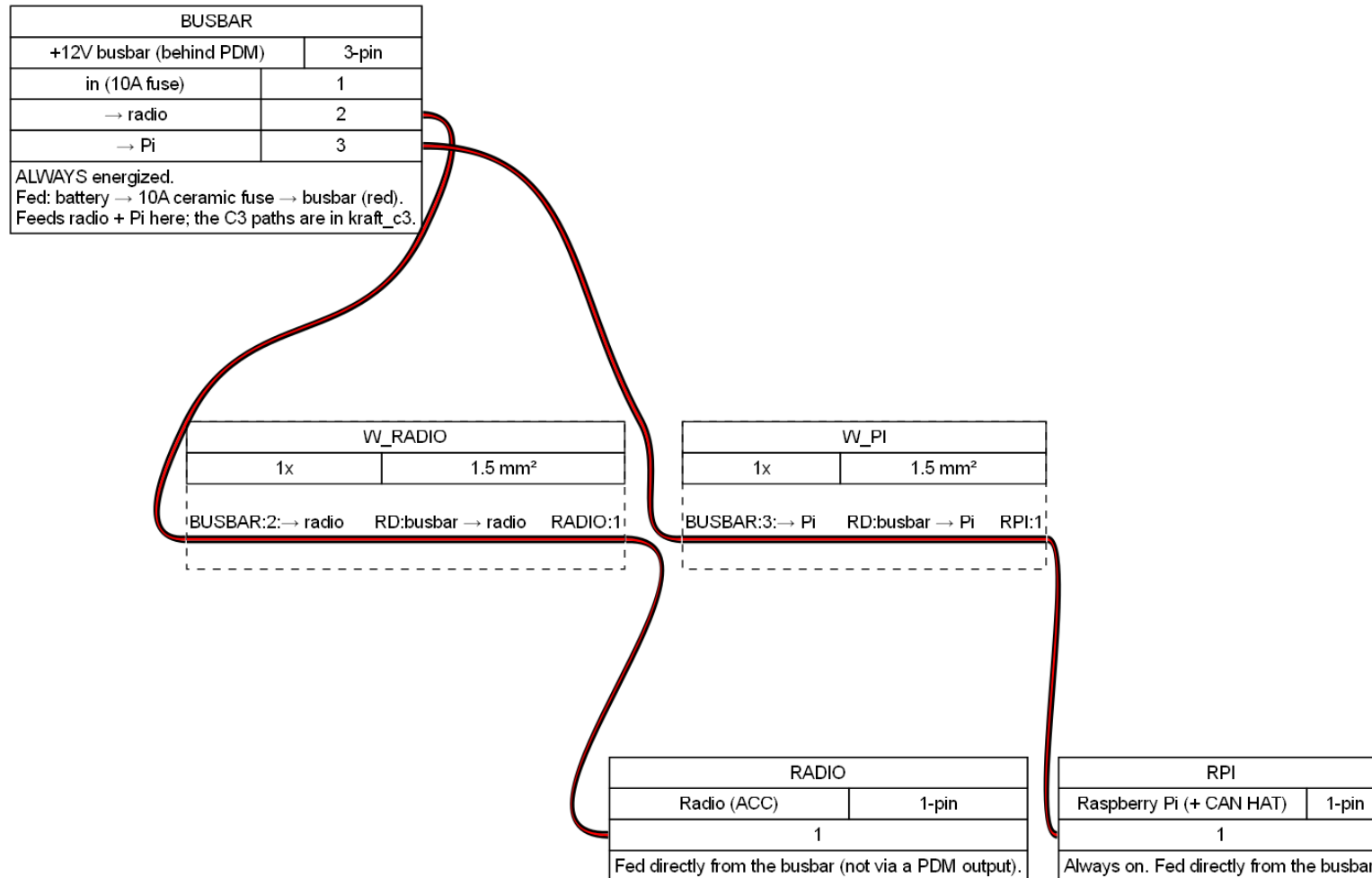


Diagram: wireviz_v8.0/2_kraft/kraft_busbar

3.2 PDM control feed C3 — ignition + hazard wake

Two paths to C3 from the busbar: normally via the ignition switch (RUN), and as a wake via the hazard switch through a blocking diode. Vertical layout.

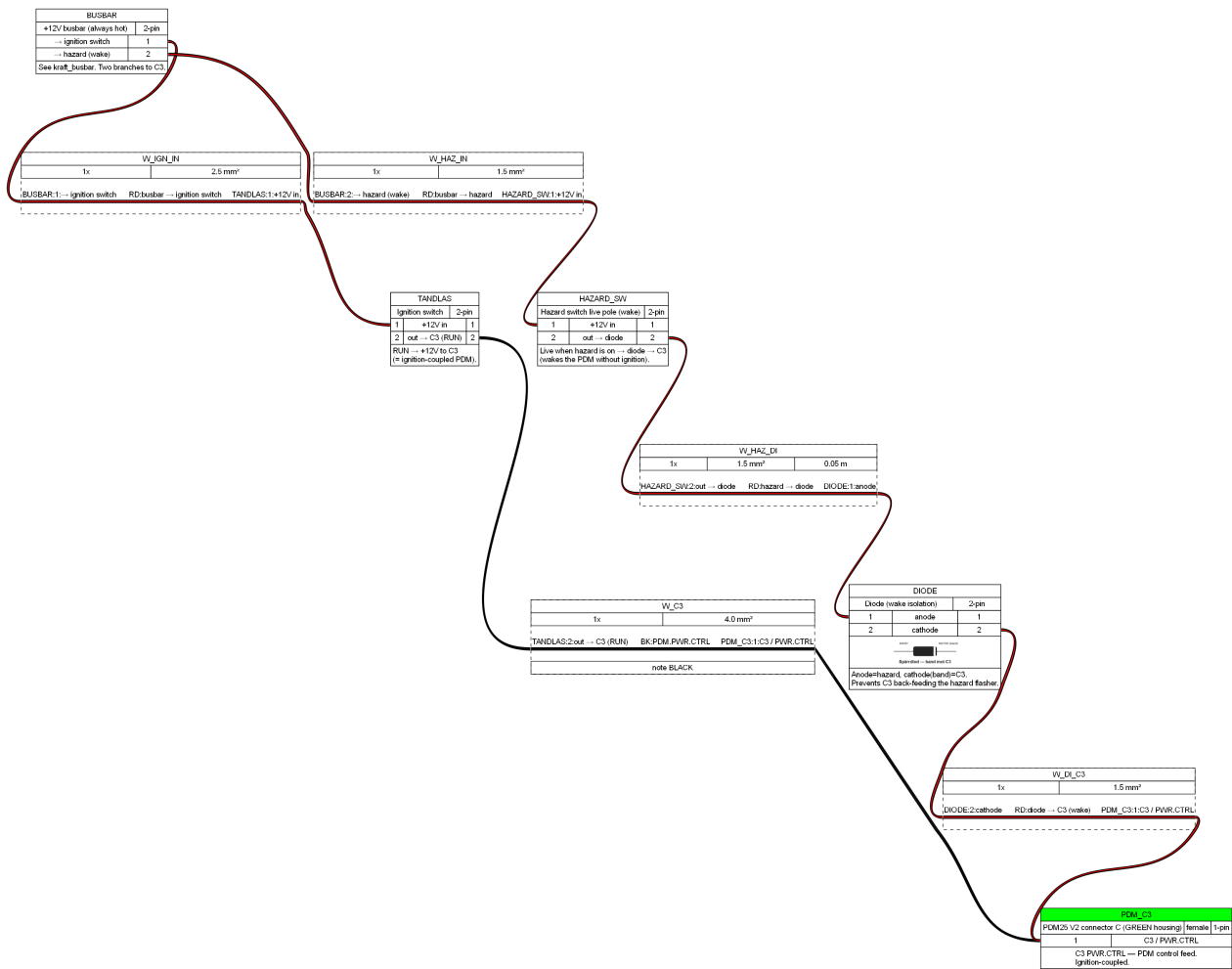


Diagram: wireviz_v8.0/2_kraft/kraft_c3

3.3 5V reference — analog inputs

C10 (5V REF, max 100 mA) feeds, via a distribution point, five SEPARATE pull-ups to the analog inputs (wiper/blower/turn mux plus coolant and fuel). Total load ~37 mA. Each input has its own pull-up — never shared.

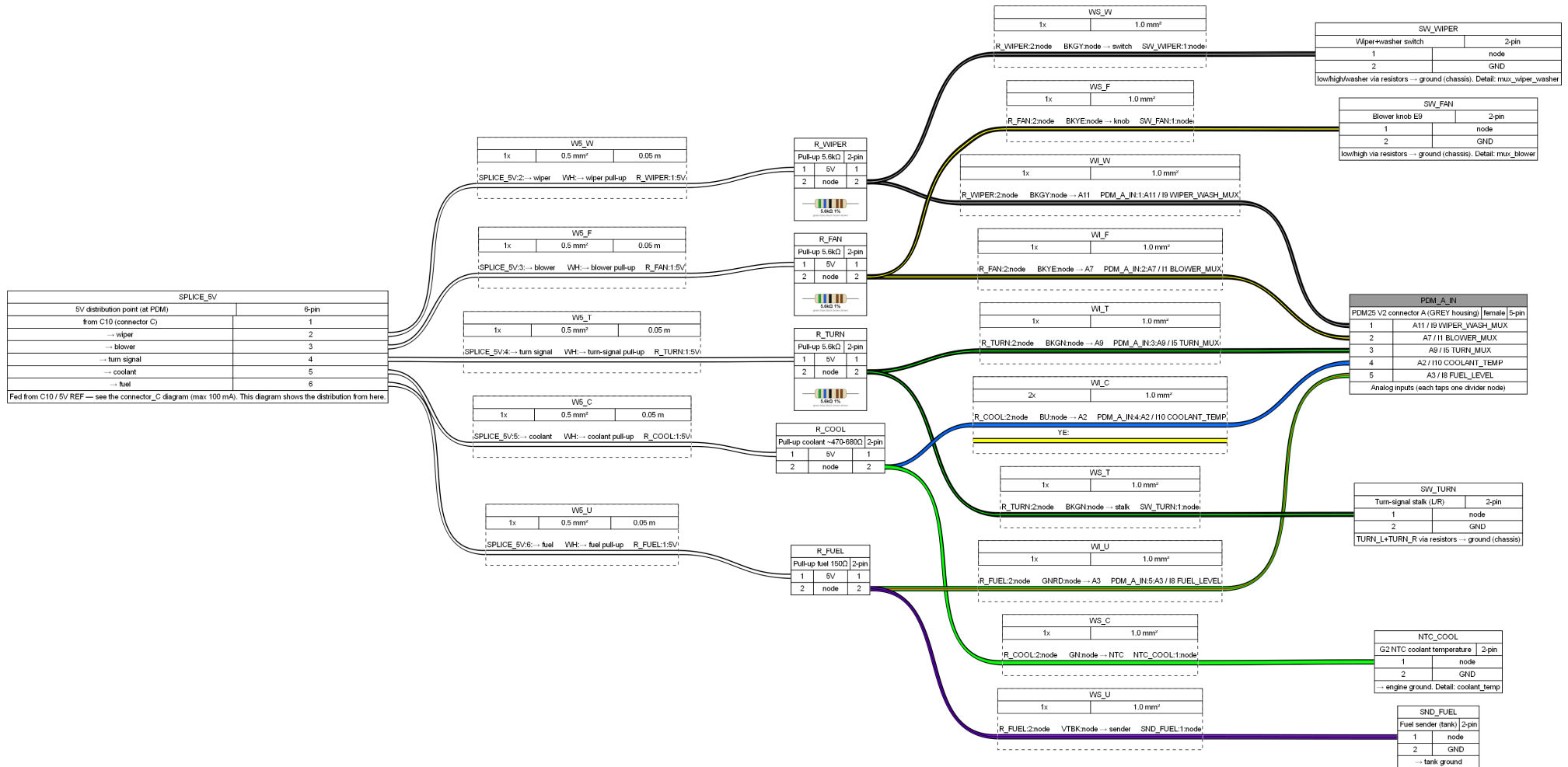


Diagram: wireviz_v8.0/2_kraft/kraft_5v

4 Circuits

4.1 Mux — wiper + washer (I9): switch + resistors

The resistor coding for the wiper+washer mux. Washer 10k, low 4.7k, high 22k on the switch outputs → chassis ground; each position gives a unique node voltage. Feed/sensing (5.6k pull-up → 5V REF) is shown in the 5V-reference diagram.

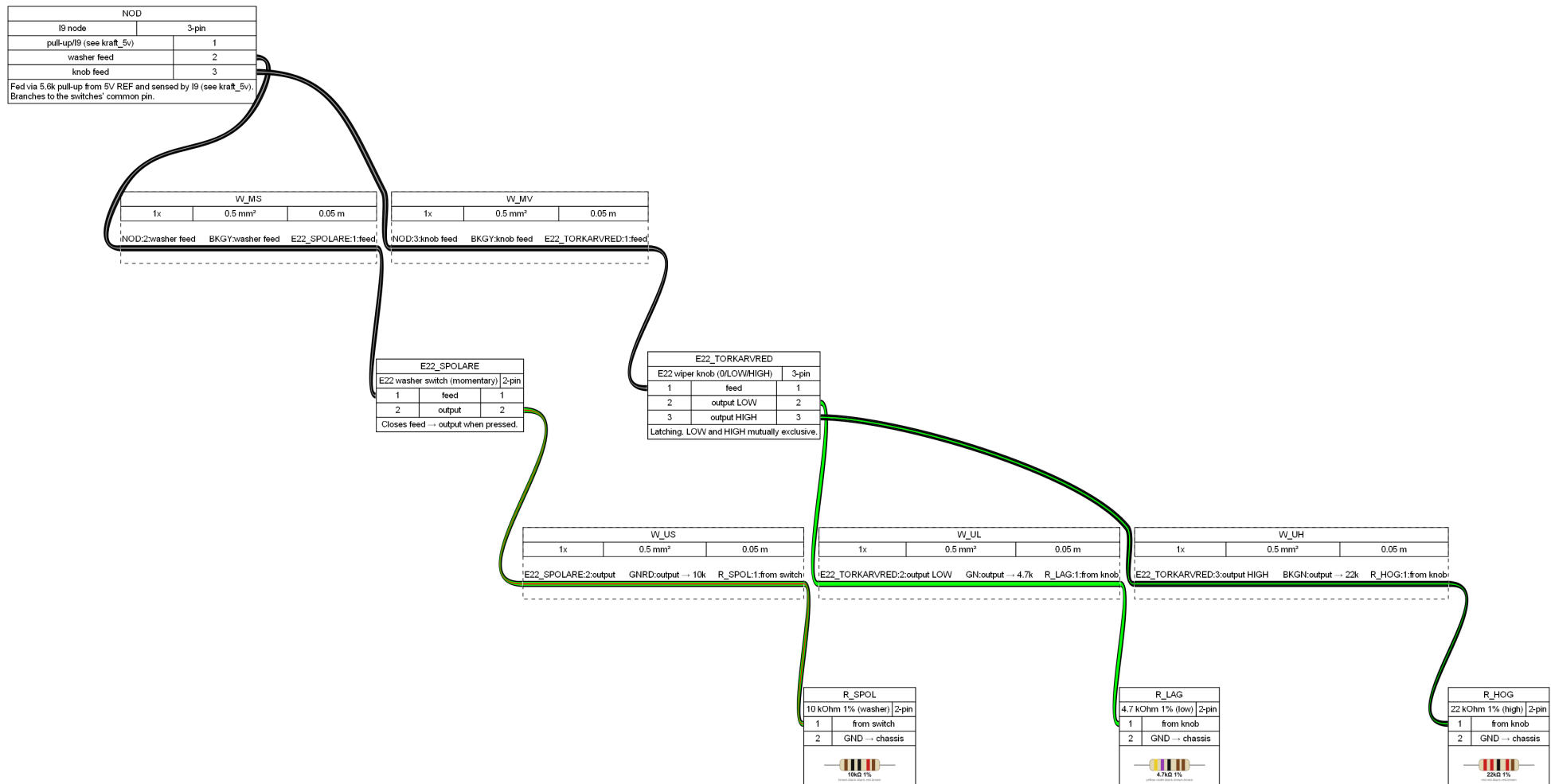


Diagram: wireviz_v8.0/3_kretsar/mux_torkare_brytare

4.2 Mux — blower low/high (I1): knob + resistors

The blower knob's resistor coding. Low 10k, high 2.2k → chassis ground. Feed via pull-up to 5V REF: see the 5V-reference diagram. Two speeds come from O11's PWM — no power resistor.

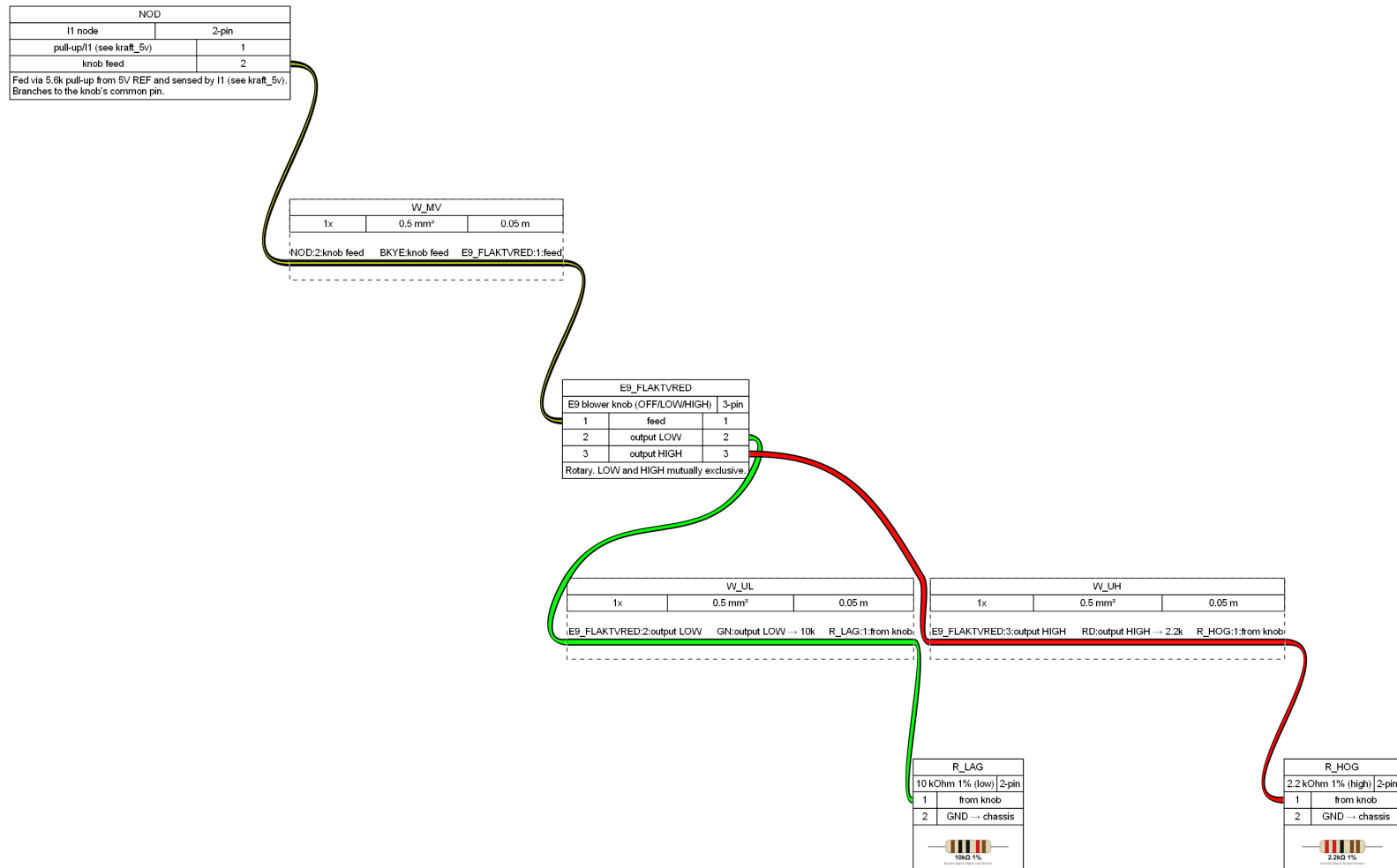


Diagram: wireviz_v8.0/3_kretsar/mux_flakt_brytare

4.3 Coolant temperature — NTC (I10)

NTC sender as a voltage divider against an external pull-up to 5V REF (~470-680Ω). High temperature = low resistance = low voltage. Ground via the sender thread in the engine block.

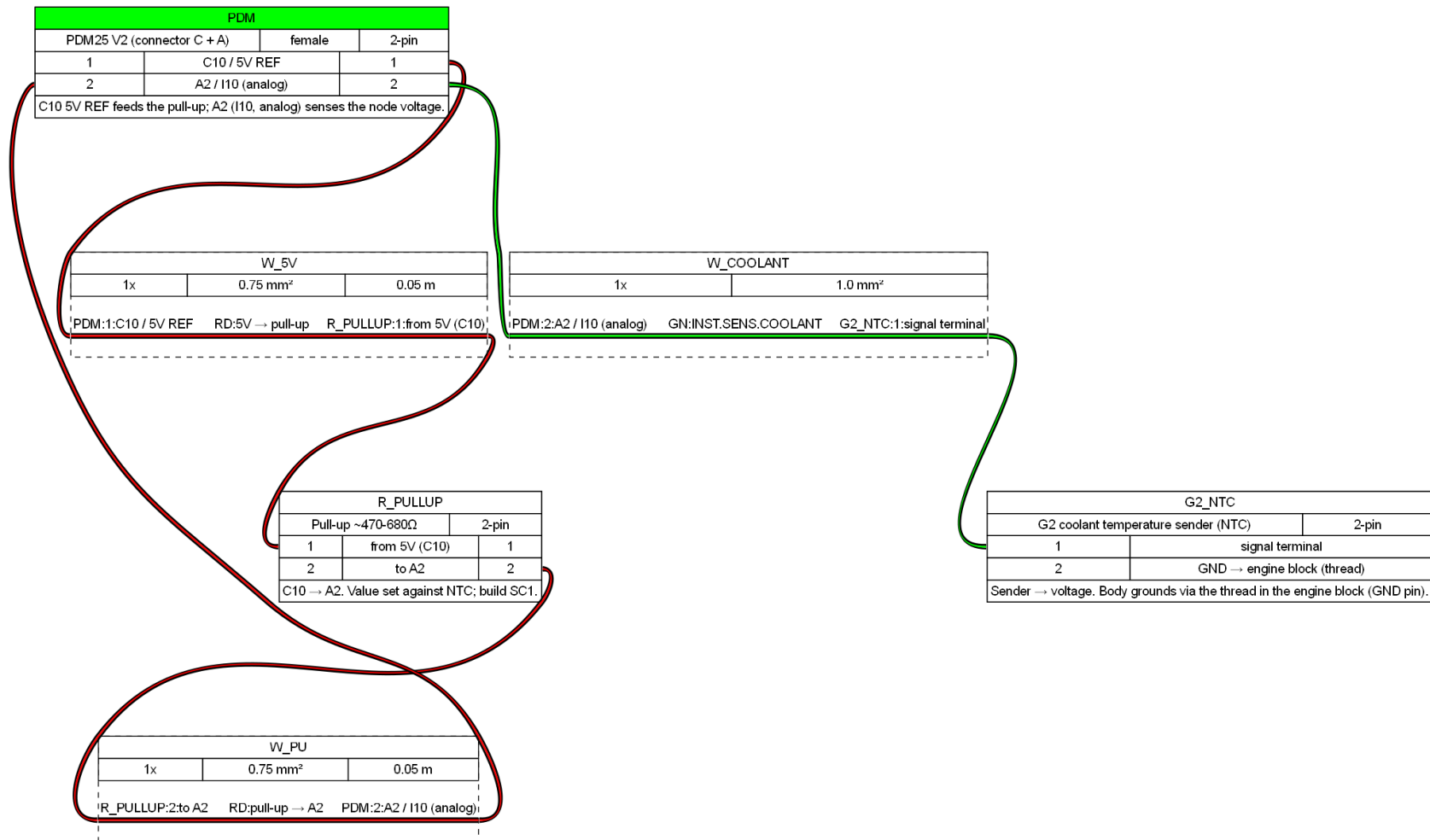


Diagram: wireviz_v8.0/3_kretsar/vattentemp

4.4 Oil-pressure switch (I14)

Simple switch-to-ground (F1). Internal pull-up to VBat. Low pressure / engine off = 0V (warning), engine running + pressure OK = high.

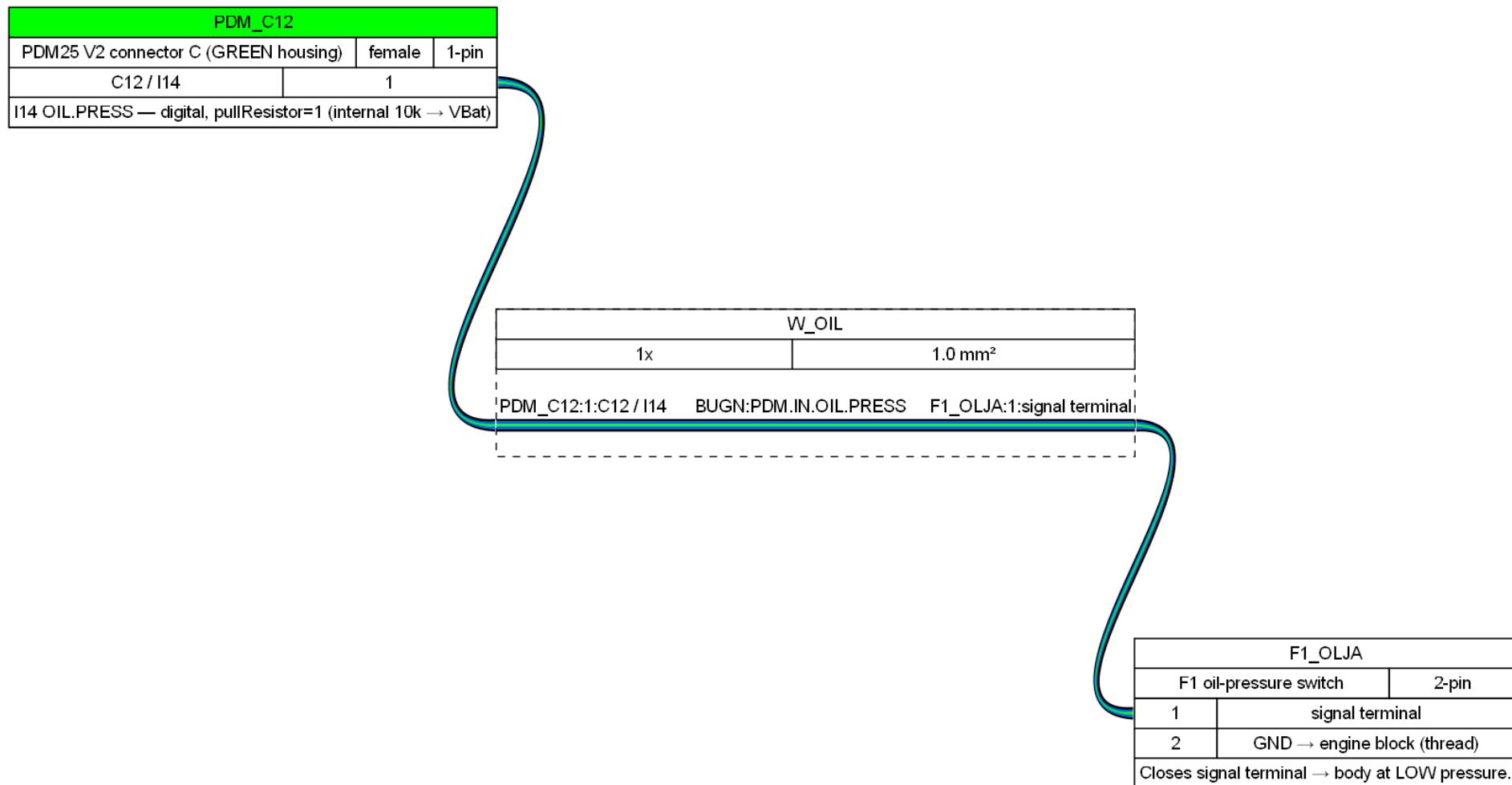


Diagram: wireviz_v8.0/3_kretsar/oljetryck

4.5 Handbrake warning (I16)

Mechanical switch-to-ground (F9), grounds when the handbrake is applied. Internal pull-up, a copy of the brake-light input.

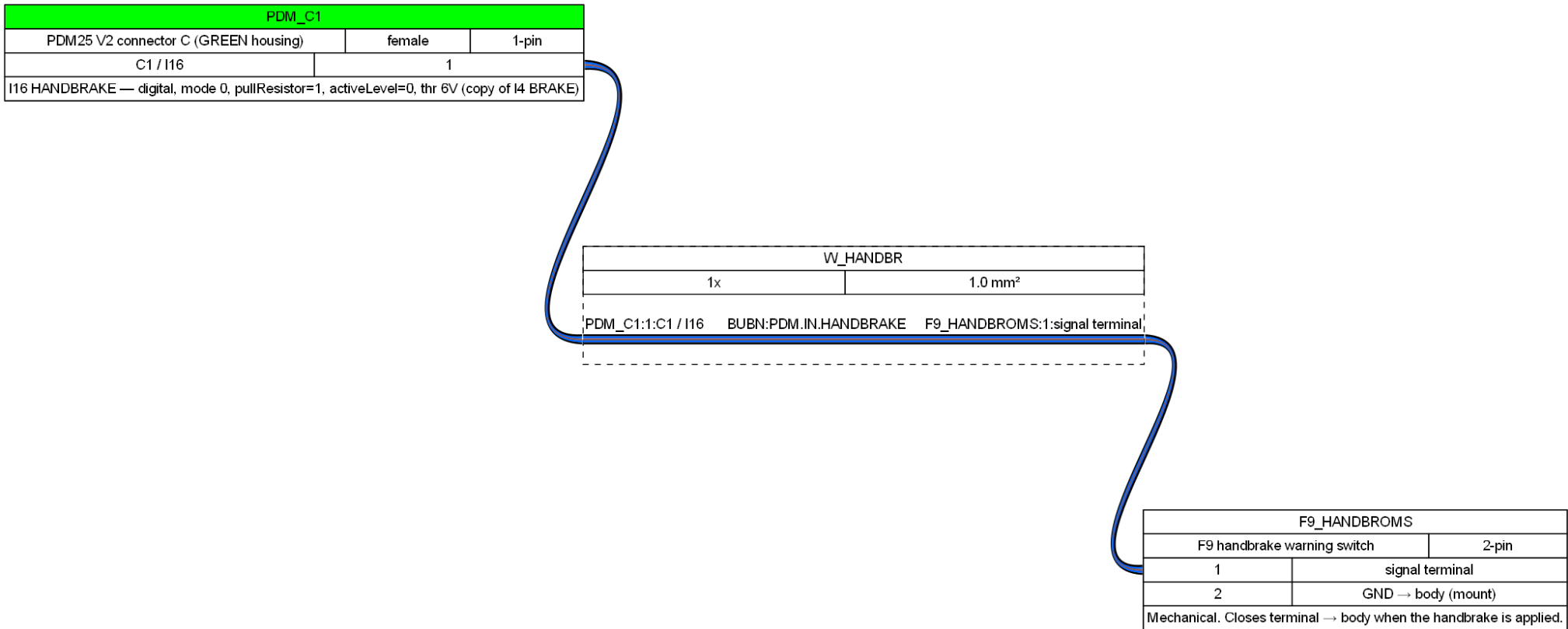


Diagram: wireviz_v8.0/3_kretsar/handbroms

4.6 Ignition coil (1/2): feed O3 → N6 → coil

Original VW, but fed by O3 instead of the ignition switch. O3 → N6 ballast resistor → coil Term. 15.

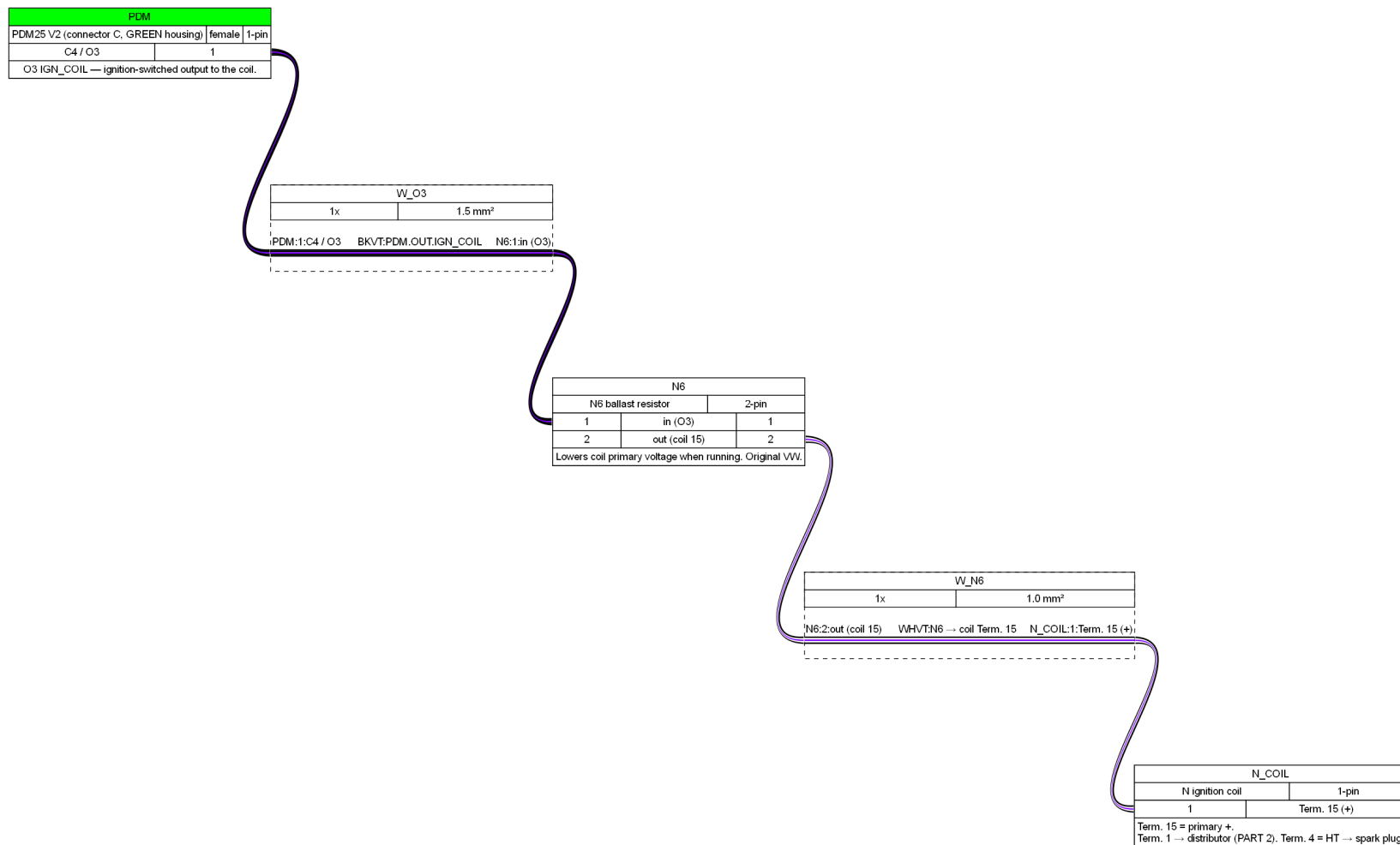


Diagram: wireviz_v8.0/3_kretsar/motor_tandspole_1_matning

4.7 Ignition coil (2/2): coil → distributor

Coil Term. 1 → distributor breaker points → ground. The points break the primary circuit → HT is induced (Term. 4 → distributor cap → spark plug, not drawn).

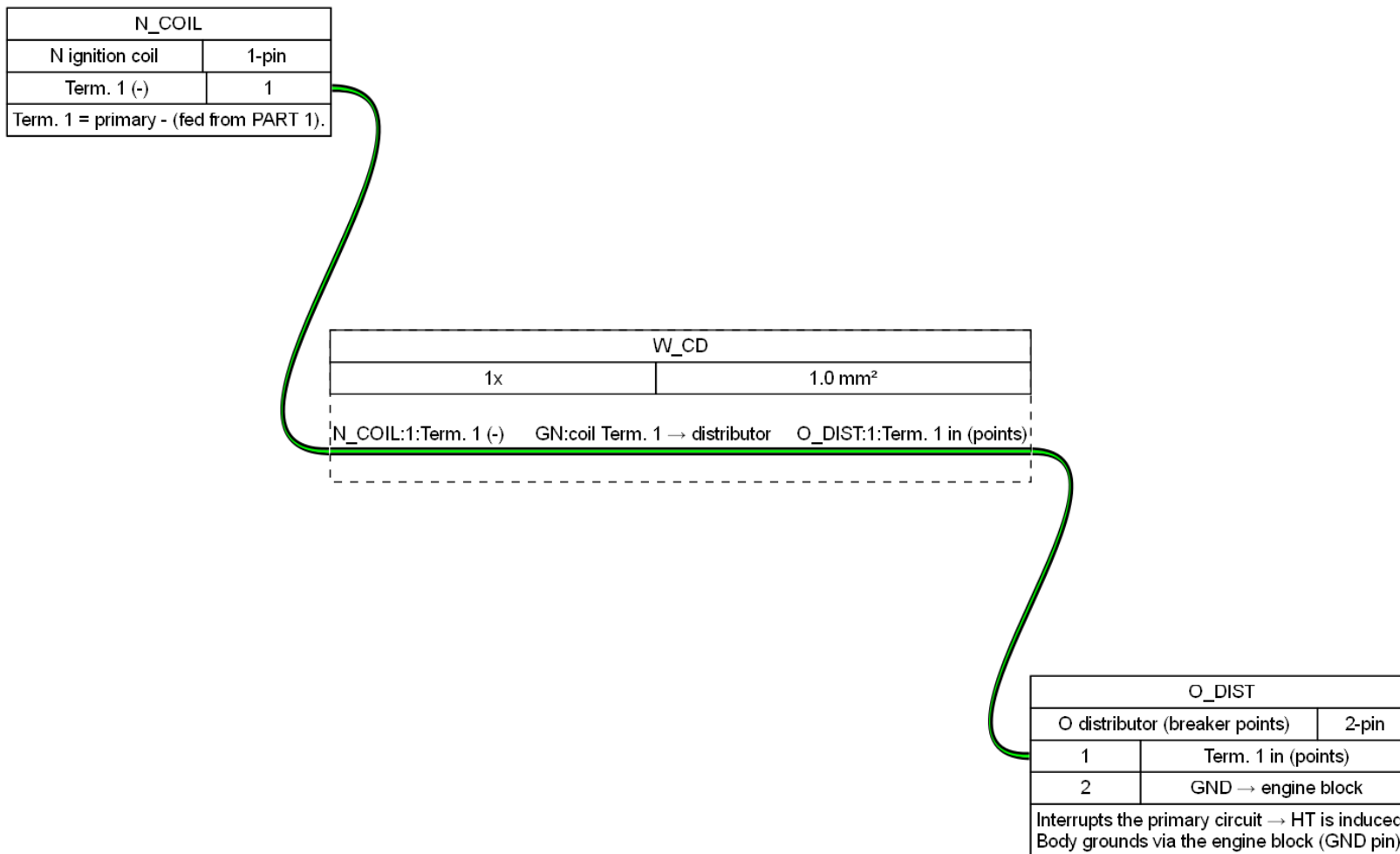


Diagram: wireviz_v8.0/3_kretsar/motor_tandspole_2_fordelare

4.8 Alternator (1/2): field excitation

O2 → 82Ω power resistor → blocking diode → alternator D+. The resistor supplies field current; the diode stops D+ (~14V) back-feeding the PDM.

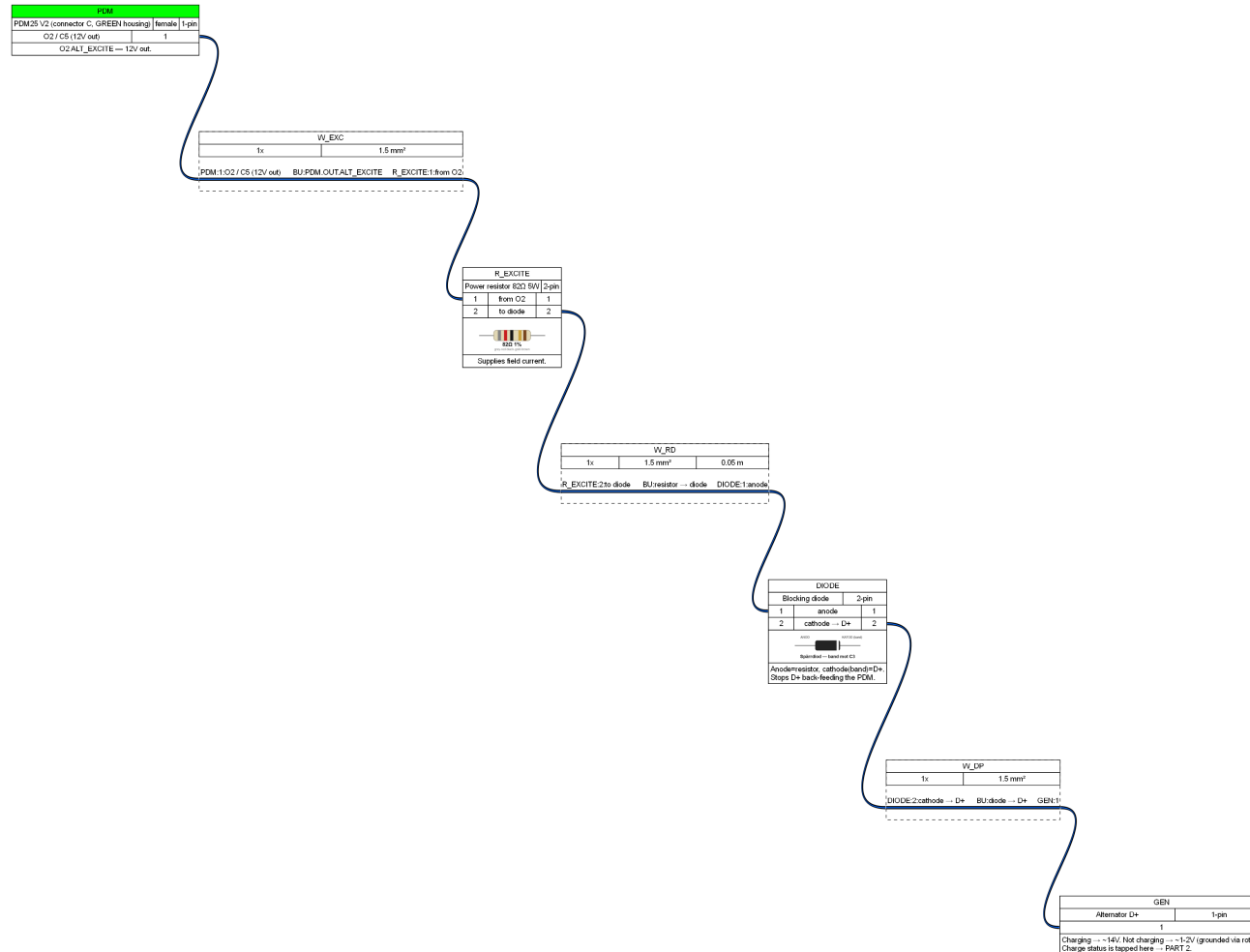


Diagram: wireviz_v8.0/3_kretsar/generator_1_excitering

4.9 Alternator (2/2): charge status

Charge status is tapped AFTER the diode to a spare PDM input (charging ~14V = high, not charging ~1V = low) → CAN → charge warning lamp in the dashboard.

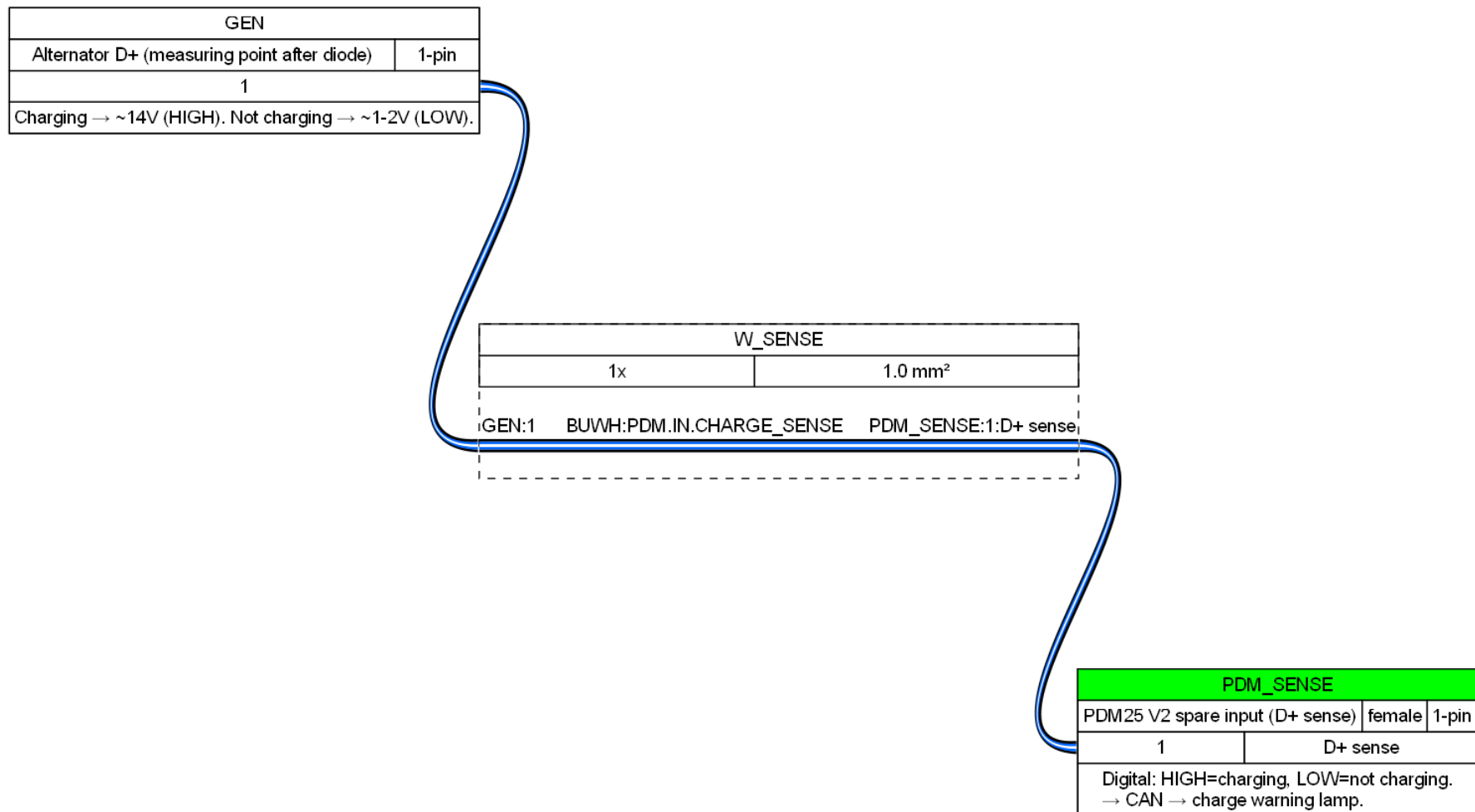


Diagram: wireviz_v8.0/3_kretsar/generator_2_laddstatus

5 Dashboard

5.1 Dashboard (1/2): Pi → GX16 rainbow ribbon

Pi 5 → GX16 10-pin ribbon in fixed color order. Canonical GPIO source: DRIVER-DASH.md §7.

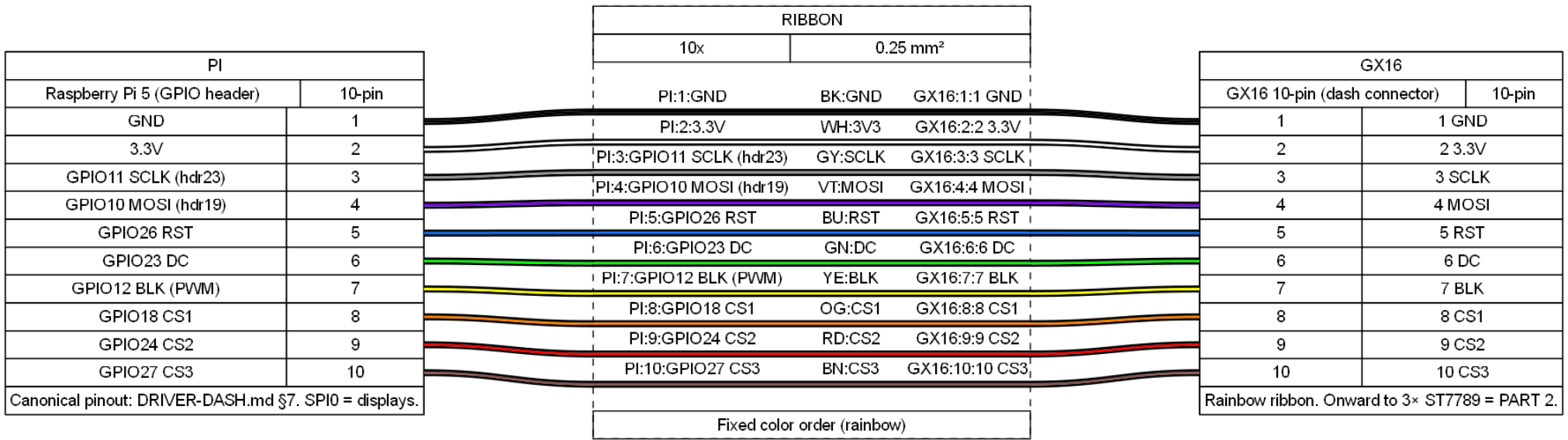


Diagram: wireviz_v8.0/4_dash/pi_1_ribbon

5.2 Dashboard (2/2): GX16 → 3× ST7789 (SPI0)

Pins 1-7 (GND/3V3/SCLK/MOSI/RST/DC/BLK) are daisy-chained to all three displays (shown as a shared bus); only CS is unique: orange→S1, red→S2, brown→S3.

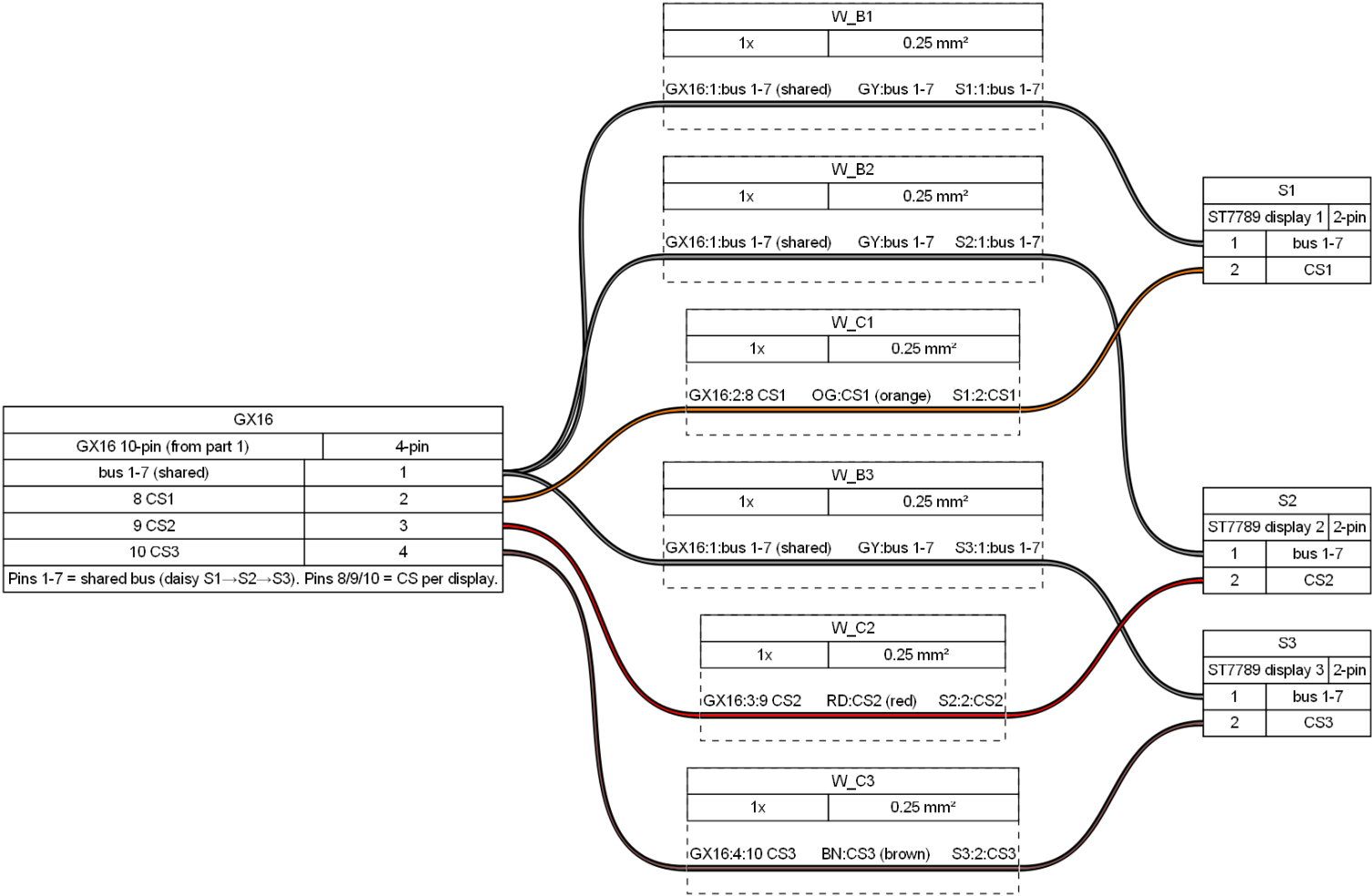


Diagram: wireviz_v8.0/4_dash/pi_2_displays

6 Appendix A — Component reference

Components on the converted vehicle, by their VW designation as used in the diagrams. From the VW LT workshop manual (Fig. 13.77, 4-cyl).

ID	Description	ID	Description
A	Battery (12 V)	H1	Horn
B	Starter motor	H2	Horn 2 (additional)
C	Alternator	L1	Headlight, left (twin-filament)
D	Ignition / start switch	L2	Headlight, right (twin-filament)
E2	Turn-signal stalk	M5	Turn signal, front left
E4	High-beam / headlight-flasher stalk	M6	Turn signal, rear left
E9	Blower (fresh-air fan) switch	M7	Turn signal, front right
E22	Wiper / washer stalk	M8	Turn signal, rear right
F	Brake-light switch	M9	Brake light, left
F1	Oil-pressure switch	M10	Brake light, right
F4	Reversing-light switch	M16	Reversing light, left
F9	Handbrake warning switch	M17	Reversing light, right
G	Fuel-gauge sender	N	Ignition coil
G2	Coolant-temperature sender	N6	Series (ballast) resistor
G6	Electric fuel pump	O	Distributor (breaker points)

7 Appendix B — Terminal designations (DIN 72552)

Standard terminal numbers used throughout the diagrams.

ID	Description	ID	Description
1	Ignition coil primary, to distributor	53	Wiper, normal speed
4	Ignition coil HT output (to distributor cap)	53a	Wiper, fast speed
15	Switched ignition + (live with ignition on)	53b	Wiper park / return
30	Battery + (permanent, unswitched)	56a	High beam
31	Ground (chassis earth return)	56b	Low beam
50	Starter control (solenoid, cranking only)	58	Parking / side lights

Terminal numbers follow the DIN 72552 standard and are common to all VW / Bosch wiring.

8 Appendix C — Earth points & grounding

In the diagrams, ground is shown as a GND pin on each consumer rather than as a drawn cable. The physical chassis earth points below are where those grounds terminate. **All ground wires are brown.**

ID	Description	ID	Description
1	Battery earth strap to body	14	Roof cross-member, passenger side
3	Engine-to-body earth strap	15	Rear, on longitudinal member
10	Instrument-panel insert	17	At steering gear
11	Behind instrument panel	18	Longitudinal member, rear left
12	Under instrument panel (former fuse-box area)	19	Longitudinal member, rear right
13	At rear door	20	In engine bay

Earth-point locations are per the VW LT factory manual for the donor vehicle and should be confirmed on the actual car. Common gauges: 0.5 mm² (lamps, senders), 1.0 mm² (motors, brake lights), 4.0 mm² and heavier (battery, engine and gearbox straps).